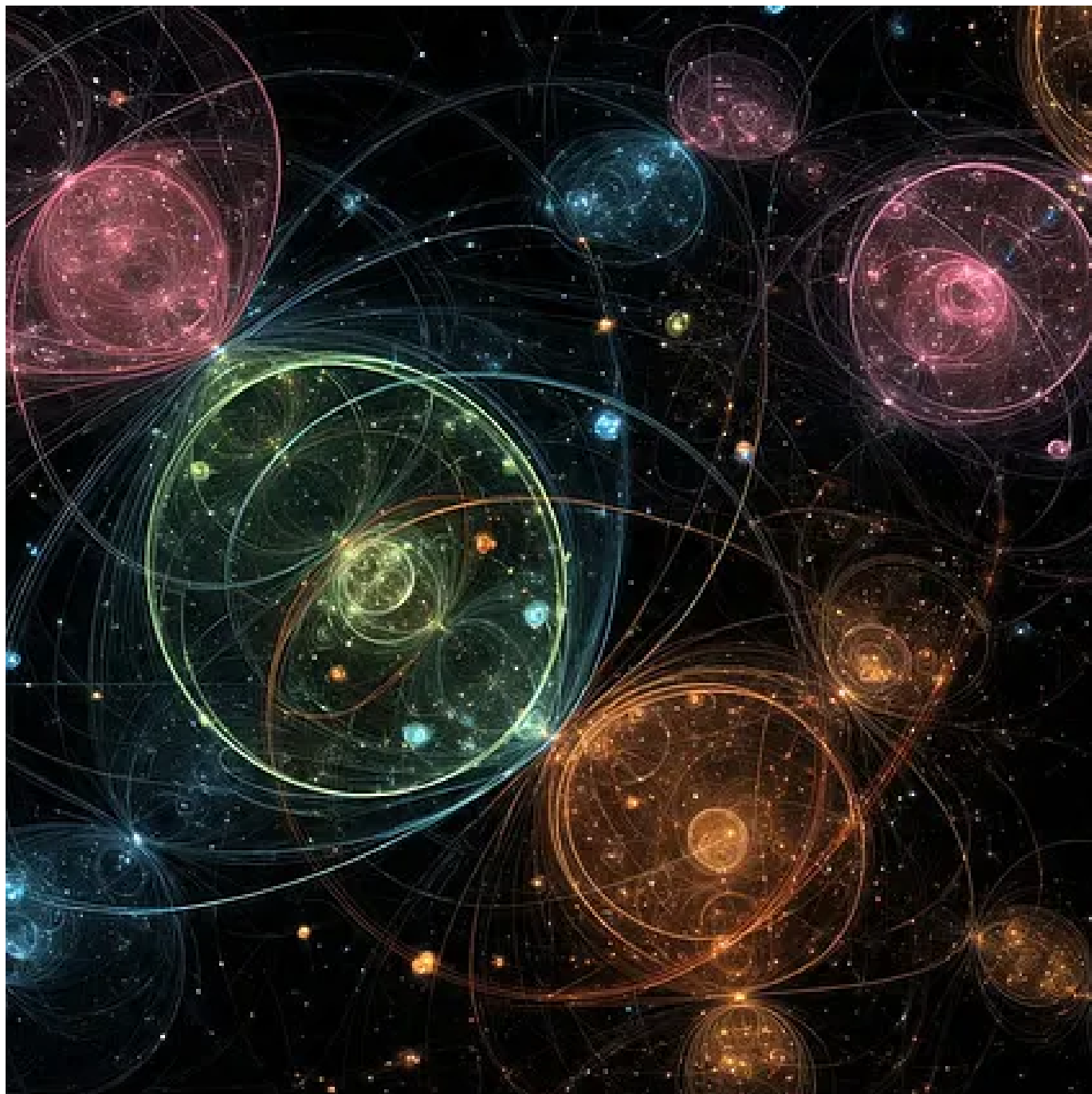


When “Embedding” Meant Delay Reconstruction, Not Word Vectors

Why Takens’ Theorem Never Needed Smooth Signals, and What NLP and ML Forgot

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Prologue: A Lost Meaning

If you search for the word “embedding” in today’s natural language processing (NLP) and machine learning (ML) literature, you’ll find static vectors. Word2Vec. GloVe. BERT embeddings. GPT projections. A word becomes a point. A sentence becomes a cloud of points. A paragraph becomes a bag of points. Meaning, in this view, is a *location* in space.

But there was another meaning of “embedding” — older, deeper, and for my money, more true to the nature of language. It came from nonlinear dynamics, not from deep learning. And it was about *trajectories*, not points. That meaning is due to Floris Takens (1981). And perhaps it’s time to bring it back.

Now Why Do I Remember?

Well that’s because I was there.

I’m Kevin R. Haylett, 63 years old. I worked as a research scientist in biomedical engineering within the UK’s National Health Service (NHS), studying complex physiological signals. And physiological signals — like speech, like the neural activations in a brain — are *nonlinear*. They are deeply, stubbornly, gloriously nonlinear. They resist linear tools. They need special methods.

During my PhD, I went to conferences. I discussed papers. I presented my own work. I even built neural networks to classify nonlinear signals ; all long before the current deep learning boom. And for me one memory stands out:

That of popping over to the Mathematics building at Manchester University and discussing my work with [Professor David Broomhead](#). You may not have heard of Professor Broomhead, but he was one of the key figures in applying dynamical systems theory to real-world data. Along with Greg King, he published the seminal 1986 paper “Extracting qualitative dynamics from experimental data” in *Physica D* — a paper that showed how to take a single time series and reconstruct the underlying attractor using singular value decomposition. That work was foundational for anyone working with experimental nonlinear data.

Professor Broomhead was genuinely surprised to see me: an engineer, and not a mathematician. He was genuinely fascinated by the way I was applying these abstract methods to real physiological signals. We had a wonderful conversation that stretched into a long hazy summer’s afternoon as we chatted over tea and biscuits. They were exciting times. I miss them.

There was as much excitement about nonlinear dynamics and chaos theory back then as there is about LLMs and AI today. And here's the connection people miss:

Language is a complex nonlinear dynamical system, rooted in a chain of measurements all the way back to the speech centres in the brain.

What Takens Actually Said

Takens' delay embedding theorem is one of the most beautiful results in dynamical systems theory. It says:

If you have a smooth dynamical system and a smooth measurement function, then a simple delay map — stacking a point with its own past measurements — reconstructs the entire hidden state space. You don't need to know the equations. You just need the sequence.

Here's what people forget: the *measurements* can be discrete. They can be integers. They can be *symbols*. The smoothness requirement applies to the *underlying system* and the *measurement function*, not to the data themselves. The data are just samples. Words are just samples. But almost every time I present my work, someone, often an LLM, and sometimes a researcher objects:

"Takens is for smooth signals, not discrete symbols."

This is wrong. Mathematically, historically, and empirically wrong. And yet it repeats, like a stuck phonograph.

The Forgotten Generation

My generation — the nonlinear dynamics generation of the 1980s and 90s — knew: discrete measurements from a continuous world still carry the full dynamical structure.

We read Packard, Crutchfield, Farmer, and Shaw (1980) — "Geometry from a Time Series" — and we realized that a single observable, properly delayed, unfolds the entire attractor. We knew that symbolic dynamics was a field for a reason. We knew that Boolean networks and cellular automata — discrete as can be — still obey the principles of state space reconstruction.

And we knew that speech, and therefore language, begins as a continuous process. Speech is continuous. Phonetics is continuous. Neural firing patterns are continuous. Language production is a continuous dynamical system that we *sample* at discrete times — into phonemes, into words. Writing compresses it further. But the dynamics are still there.

Mainstream NLP threw that away. It chose static vectors. It chose points. It chose geometry without time.

What's the Point?

So where does this lead? What's the point?

The point is this: if we follow the route of nonlinear dynamical systems theory, we can design and implement Takens' method of delays to create a new type of fully working language model.

We can build a Takens-based Transformer. This isn't mystical and it doesn't need "attention" as the transformer defines it. It doesn't need billion-parameter static embeddings. It needs one thing: Takens' embedding theorem. And, for me, importantly it works. How do I know? Because I built it, yes, a fully functioning language model based on Takens' theorem: a Takens-based Transformer.

No classical word vectors. No static points pretending to carry meaning. Just trajectories: curved, winding, interacting, unfolding in a reconstructed state space where meaning lives in the dynamics, not in a location.

You can see it here, with a formal paper, the underlying explanation of why it works and the link to GitHub for the code:

👉 <https://www.finitemechanics.com/takens-transformer/index.html>

Why This Matters for NLP and ML

Perhaps static embeddings are a convenience, not a final truth. They compress a temporal sequence (a sentence) into a fixed point. Two different sentences that end up near the same

point can be treated as similar ; even if their *paths* through semantic space were completely different.

However, language is about *paths*. It's about how a meaning unfolds. It's about the curve of a clause, the arc of a narrative, the winding of a conversation. You cannot capture that with a point Takens trajectories can.

Moreover, the entire objection that “Takens requires smooth signals” has frozen innovation in this area. It's perhaps best seen as a myth that belongs to a world of make-believe and not of practical real world application and engineering.

A Challenge to the Reader

Next time someone tells you that Takens' theorem only works on smooth, real-valued data, ask them:

- Did you read the original Takens paper?
- Where does it say data must be real-valued?
- Does symbolic dynamics not exist?
- Are Boolean networks and cellular automata not dynamical systems?
- Did Broomhead, King, and the Manchester school not apply these methods to discrete time series?

This article addresses these questions. Haylett, K. R. (2026). Takens' Theorem Applies to Discrete Symbol Sequences.

Coda: Remembering

When I started in nonlinear dynamics, “embedding” was a verb. It meant unfolding a trajectory from its shadows. It meant recovering the hidden attractor. It meant *understanding the dynamics*.

Somewhere along the way, NLP and the ML community borrowed the noun and forgot the verb.

I remember popping over to the Mathematics building at Manchester University. I remember Professor David Broomhead's surprise and fascination — a giant in the field who took the time to talk with an enthusiastic engineer. I remember the excitement.

This article is a small attempt to remember and to build something new from something old. And this article is dedicated to Professor Broomhead, who is sadly no longer with us and who, on one sunny afternoon with a look of surprise and joy, was delighted to hear about my work ; quite unlike those in the ML community who appear to be locked into a world of static vectors.



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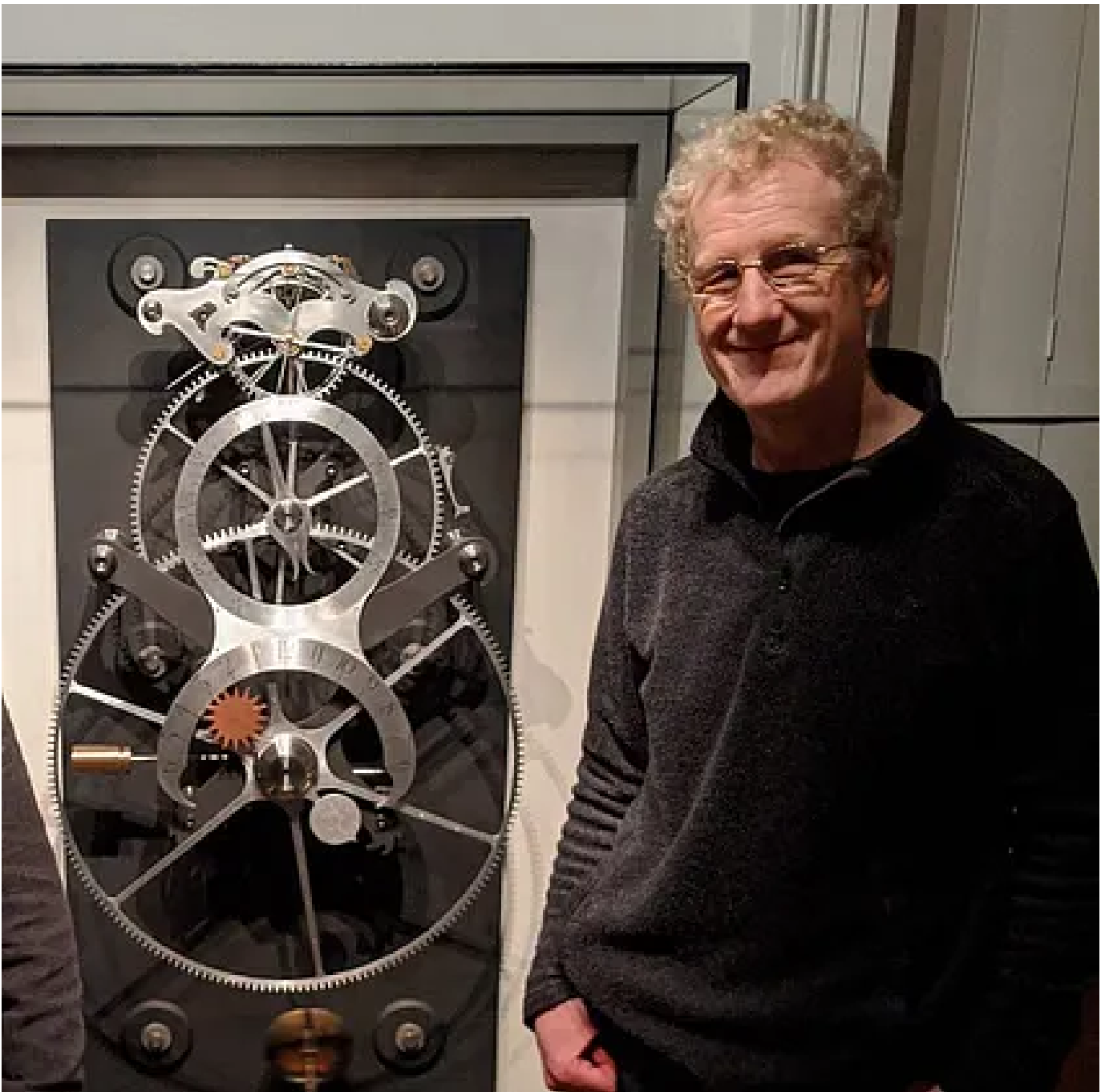
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- Haylett, K. R. (2026). Takens' Theorem Applies to Discrete Symbol Sequences. [[Link](#)]

Further Reading

[An article on language as a nonlinear dynamical system](#)

[A collection of papers, essays and, monographs](#)

[Website with further works](#)



Standing next to the Burgess B Clock at the Greenwich Royal Observatory. This is the most accurate clock with a pendulum. It keeps to within 0.6 Seconds in 100 days.

About the Author

Kevin R. Haylett is a 63-year-old former biomedical engineering research scientist in the UK's NHS. He worked on complex physiological signals, neural networks for nonlinear classification, and delay embedding methods. He remembers popping over to Manchester University's Mathematics building to discuss his work with Professor David Broomhead, to

whom this work is dedicated. He built a Takens-based language model because he remembered what “embedding” used to mean. He can be found [here](#).

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Omne quod est, finitum est; tantum per mensuram cognosci potest

Everything that exists is finite; it can only be known by measure

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